

Haar limits of optimal wild cycles

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Abstract

For a complete algebraically closed ultrametric field of odd characteristic p , we prove that the uniform measures on the small cycles of $P_\lambda(z) = \lambda z + z^2$ converge whenever $0 < |\lambda - 1| < 1$. The conclusion concerns the entire sequence of cycles of least periods p^e , over every field and multiplier in this range. Their union has compact closure in the classical field, and their Hausdorff limit is a compact invariant set on which P_λ is conjugate to addition by one on $\mathbb{Z}_p$. The limiting probability is the corresponding nonatomic Haar measure, with support consisting entirely of type-I points. The main estimate is an exact average-contact identity between any fixed cycle and every higher cycle. With $s = \lambda - 1$ and $r = (p - 1)/p$, its normalized average c_d satisfies $c_d \geq dr^3 + r^2(1 + 1/p)$; the Hausdorff and essential-supremum transport errors are at most $|s|^{c_d}$. A finite upper bound on contacts with each old cycle excludes periodic points from the limit. The proof answers the selected-cycle convergence question posed by Lindahl and Rivera-Letelier without a discrete-valuation or field-nesting assumption.

1 Introduction and main theorem

The location of a periodic orbit does not determine how a sequence of such orbits is distributed. This distinction is particularly concrete for the polynomial $P_\lambda(z) = \lambda z + z^2$ in positive characteristic: its small cycles occupy one sphere, although their ordinary periods tend to infinity. We prove that these cycles converge to a classical compact adding-machine system, and give a quantitative estimate for their supports and uniform probability measures.

Throughout the article, p is an odd prime, $(K, |\cdot|)$ is a complete algebraically closed ultrametric field of characteristic p , and

$$s = \lambda - 1, \quad 0 < |s| < 1, \quad P(z) = (1 + s)z + z^2, \quad r = \frac{p - 1}{p}. \quad (1.1)$$

We normalize the real valuation by $v(x) = \log |x| / \log |s|$ for $x \neq 0$ and $v(0) = +\infty$. For $e \geq 1$, let Π_e be the unique cycle in the open unit disk of ordinary least period p^e , and put

$$\mu_e = p^{-e} \sum_{\alpha \in \Pi_e} \delta_\alpha. \quad (1.2)$$

The existence, uniqueness, distinctness, and common radius $|\alpha| = |s|^r$ are the optimal-cycle results of Lindahl–Rivera-Letelier [4, Theorem C and the $q = 1$ discussion]. All masses in (1.2) are real numbers.

Distances between classical points are $d(x, y) = |x - y|$. For nonempty compact sets E, F we use the Hausdorff distance

$$d_H(E, F) = \max \left\{ \sup_{x \in E} \inf_{y \in F} d(x, y), \sup_{y \in F} \inf_{x \in E} d(x, y) \right\}.$$

A coupling of probabilities σ, τ is a probability on the product with those marginals. Write $W_1(\sigma, \tau)$ for the infimum of $\int d(x, y) d\pi$ over couplings, and $W_\infty(\sigma, \tau)$ for the infimum of their essential supremum of $d(x, y)$. All measures used in these distances below are carried by a common compact metric space. Weak convergence on $\mathbb{P}_K^{1, \text{an}}$ means convergence against every continuous real-valued function there.

Theorem 1.1 (Compact adding-machine limit of the optimal cycles). *Under (1.1), the following conclusions hold.*

(i) *The classical closure*

$$C = \text{cl}_K \left(\bigcup_{e \geq 1} \Pi_e \right)$$

is compact. The entire sequence Π_e has a nonempty Hausdorff limit $\mathcal{A} \subset C$, with $P(\mathcal{A}) = \mathcal{A}$ and $|a| = |s|^r$ for $a \in \mathcal{A}$.

(ii) *For $d \geq 1$ and any $\beta \in \Pi_d$, the number*

$$c_d = \frac{p-1}{p^{d+1}} v \left((P^{p^d})'(\beta) - 1 \right) \quad (1.3)$$

is finite and independent of β . For every $e > d$,

$$\frac{1}{p^e} \sum_{\alpha \in \Pi_e} v(\beta - \alpha) = c_d \geq dr^3 + r^2 \left(1 + \frac{1}{p} \right). \quad (1.4)$$

(iii) *The entire sequence μ_e converges weakly on $\mathbb{P}_K^{1, \text{an}}$ to a probability μ supported exactly on $\mathcal{A}$. For each $d \geq 1$,*

$$d_H(\Pi_d, \mathcal{A}) \leq |s|^{c_d}, \quad W_1(\mu_d, \mu) \leq W_\infty(\mu_d, \mu) \leq |s|^{c_d}. \quad (1.5)$$

(iv) *There is a homeomorphism $\theta : \mathbb{Z}_p \rightarrow \mathcal{A}$ such that $P(\theta(t)) = \theta(t+1)$. The measure μ is the pushforward of Haar probability under θ , is nonatomic, and is the unique P -invariant probability on $\mathcal{A}$. In particular $\mathcal{A}$ has no periodic points and consists entirely of type-I points of $\mathbb{P}_K^{1, \text{an}}$.*

This answers Problem 1.3 in the 26 May 2015 version of Lindahl–Rivera-Letelier [4], with its original field and multiplier quantifiers. The assertion concerns each fixed allowed λ ; it does not give convergence uniform as $|\lambda - 1| \rightarrow 1$. The set C contains the old finite cycles and their invariant atomic measures. Unique ergodicity in the theorem is asserted only on $\mathcal{A}$, not on C .

The main arithmetic step is (1.4). Differentiating a small-cycle factorization at a level- d anchor expresses its multiplier through within-cycle distances. Differentiating every higher return quotient at the same anchor then gives its mean contact with Π_e . A characteristic- p displacement estimate makes that mean diverge as d grows. Because P is isometric on the relevant disk, a single close pair couples the complete cycles. The finite value of the same mean also separates every fixed old cycle from the limit, which is the additional input needed for aperiodicity.

Relation to other measures and classical inputs. Jacobs proves convergence of Rumely's crucial measures to the canonical dynamical measure [3, Theorem 2]. His reduction-dependent weights are supported at type-II points; they are not the uniform measures (1.2). This distinction is one of measures, not of characteristic. The periodic-point norm bounds of Nordqvist–Rivera–Letelier [5, Theorem 3] concern ramified multiplier-one series and do not provide the cross-level estimate used here. We use classical seminorm topology for the Berkovich line [1]. The inverse-limit description and invariant measure of compact minimal equicontinuous systems are also classical; see Hurder–Lukina [2, author version, Sections 2.2–2.3]. We give the required cyclic-partition proof, including the finite alternative, in full.

The argument is independent of any full-inertia theorem for local periodic fields, of Artin–Schreier character stabilization, or of a nested sequence of splitting fields. The coefficient-field construction in Section 2 is a proof device inside arbitrary K , not a restriction of the theorem to a Laurent-series field. Sections 3–4 prove the contact estimate; Sections 5–6 prove the compact, measure, and adding-machine conclusions.

2 Small factors over an arbitrary complete field

We first construct the factorization used in the contact calculation without assuming irreducibility or local inertia. Write

$$F_e(z) = P^{\circ p^e}(z) - z \quad (e \geq 0), \quad Q_e(z) = F_e(z)/F_{e-1}(z) \quad (e \geq 1). \quad (2.1)$$

These indices label p -power returns; the superscript $\circ$ always denotes ordinary iteration. The minimal ramification of $g(z) = z + z^2$ gives the source input

$$\text{ord}_z(g^{\circ p^e}(z) - z) = 1 + \frac{p^{e+1} - 1}{p - 1} \quad (e \geq 0). \quad (2.2)$$

This is the $q = 1$ reduction underlying the cited optimal-cycle theorem. The other input is the already established set of p^e distinct points Π_e , each of valuation r . We will use degree to identify the whole small factor, so no further separability premise is needed.

Lemma 2.1 (Coefficient-field realization). *Let $k_0 \subset K$ be the subfield of elements algebraic over $\mathbb{F}_p$. There is an isometric embedding*

$$\iota : k_0((t)) \longrightarrow K, \quad t \longmapsto s,$$

where $k_0((t))$ has norm $|s|^{\text{ord}_t}$.

Proof. Algebraic closedness of K makes k_0 an algebraic closure of $\mathbb{F}_p$. Every nonzero element of k_0 lies in a finite field and has norm one, since it is a root of unity. For a Laurent series $a(t) = \sum_{n \geq n_0} a_n t^n$, set

$$\iota(a) = \sum_{n \geq n_0} a_n s^n.$$

Its terms tend to zero, so the ultrametric inequality and completeness of K give convergence. If n_* is its first nonzero coefficient index, the leading term has norm $|s|^{n_*}$ and the remaining tail has norm at most $|s|^{n_*+1}$. Thus

$$|\iota(a)| = |s|^{n_*} \quad (a \neq 0). \quad (2.3)$$

Addition and multiplication are preserved on finite Laurent polynomials. Approximating arbitrary series by finite truncations and using continuity of both operations proves the same for series.

Equation (2.3) gives injectivity, and preservation of 1 and multiplication gives preservation of inverses. The norm identity proves the isometry. No equality between the residue field of K and k_0 is assumed. $\square$

Lemma 2.2 (Canonical small factor). *For every $e \geq 1$ there are monic $M_e, V_e \in K[z]$ with integral coefficients such that*

$$F_e = F_{e-1}M_eV_e, \quad M_e(z) = \prod_{\alpha \in \Pi_e} (z - \alpha), \quad v(V_e(x)) = 0 \quad \text{if } v(x) > 0. \quad (2.4)$$

Proof. Work first over the complete discrete valuation ring $R = k_0[[t]]$ with $P_t(z) = (1+t)z + z^2$, and attach a subscript t to the returns in (2.1). For a polynomial h , the relation $h(z) = z$ in the quotient by $h(z) - z$ implies $h^{\circ p}(z) = z$ there. Taking $h = P_t^{\circ p^{e-1}}$ proves divisibility of $F_{e,t}$ by $F_{e-1,t}$. Monic division gives a monic quotient $Q_{e,t}$ in $R[z]$, and reduction commutes with this division.

Successive orders in (2.2) differ by p^e . Consequently

$$\overline{Q}_{e,t}(z) = z^{p^e} \overline{V}_e(z), \quad \overline{V}_e(0) \neq 0.$$

The displayed monic factors are coprime. Coprime Hensel factorization over R gives monic polynomials $M_{e,t}, V_{e,t} \in R[z]$ with

$$Q_{e,t} = M_{e,t}V_{e,t}, \quad \deg M_{e,t} = p^e, \quad \overline{M}_{e,t} = z^{p^e}, \quad \overline{V}_{e,t}(0) \neq 0.$$

Apply Lemma 2.1 coefficientwise. The resulting M_e, V_e have integral coefficients, and $V_e(0)$ is a unit. At an input of positive valuation, every positive-degree term of V_e has positive valuation. Hence $v(V_e(x)) = 0$ there.

For $\alpha \in \Pi_e$, its least period implies $F_{e-1}(\alpha) \neq 0$. The transported factorization therefore forces $M_e(\alpha) = 0$. There are p^e distinct such points and M_e is monic of degree p^e . This proves its exact root product in (2.4) and completes the argument. $\square$

The construction takes place over one coefficient field that embeds in every allowed K . It uses neither a generic-specialization argument nor any conclusion about the Galois group of M_e .

3 Native isometry and displacement bounds

Set $\mathcal{D} = \{x \in K : v(x) \geq r\}$, a complete closed disk. It contains every Π_e . Since $|1+s| = 1$ and $|x| < 1$ on this disk, $P(\mathcal{D}) \subseteq \mathcal{D}$. The identity

$$P(x) - P(y) = (x - y)(1 + s + x + y) \quad (3.1)$$

contains slightly more information than isometry: the unit factor has residue one.

Lemma 3.1 (Residue-one difference quotients). *For $x, y \in \mathcal{D}$ and every integer $a \geq 0$,*

$$|P^{\circ a}(x) - P^{\circ a}(y)| = |x - y|.$$

If $x \neq y$, then

$$\frac{P^{\circ a}(x) - P^{\circ a}(y)}{x - y} \in 1 + \{u \in K : v(u) \geq r\}. \quad (3.2)$$

Proof. In (3.1), $v(s) = 1 > r$ and $v(x), v(y) \geq r$. Thus the second factor lies in the set on the right of (3.2) and has norm one. Products of elements of that set remain in the set. Iteration proves both assertions. $\square$

Fix $d \geq 1$ and $\beta \in \Pi_d$. For $0 \leq j < d$, define the finite quantities

$$\delta_j = v(P^{\circ p^j}(\beta) - \beta). \quad (3.3)$$

Lemma 3.2 (Displacements indexed by native time). *For $1 \leq a < p^d$,*

$$v(P^{\circ a}(\beta) - \beta) = \delta_{v_p(a)}. \quad (3.4)$$

There are $(p-1)p^{d-j-1}$ indices contributing the value δ_j in this formula. Distinct j need not give distinct valuation values.

Proof. Write $a = kp^j$ with $p \nmid k$ and set $G = P^{\circ p^j}$. The telescoping identity

$$G^{\circ k}(\beta) - \beta = \sum_{i=0}^{k-1} (G^{\circ(i+1)}(\beta) - G^{\circ i}(\beta))$$

may be divided by $G(\beta) - \beta \neq 0$. By (3.2), each resulting summand has residue one. Their sum has residue $k \neq 0$, so it has valuation zero. This proves (3.4). The count is the number of integers in $\{1, \dots, p^d - 1\}$ divisible by p^j but not by p^{j+1} . $\square$

Lemma 3.3 (Exponential displacement bound). *For all $d \geq 1$, $\beta \in \Pi_d$, and $0 \leq j < d$,*

$$\delta_j \geq (p^j + 1)r. \quad (3.5)$$

Proof. For $H(z) = \sum_i h_i z^i \in K[z]$, define the weighted Gauss valuation

$$w_r(H) = \min_{h_i \neq 0} \{v(h_i) + ir\}, \quad w_r(0) = +\infty.$$

Let U be the K -linear operator $UH = H \circ P$, and put $\Delta = U - I$. For every integer $i \geq 1$,

$$\Delta(z^i) = z^i((1 + s + z)^i - 1).$$

The constant coefficient in the bracket has valuation at least one if nonzero. Every positive-degree term has weighted valuation at least r , since its coefficient is integral. Because $1 > r$, the bracket has weight at least r . Constants have zero image under Δ , and the ultrametric inequality gives

$$w_r(\Delta H) \geq w_r(H) + r. \quad (3.6)$$

It follows by induction that $w_r(\Delta^N z) \geq (N+1)r$ for all positive integers N . The operators U and I commute, and their coefficient field has characteristic p . Thus

$$\Delta^{p^j} = U^{p^j} - I.$$

Evaluation at β , whose valuation is r , cannot decrease the weighted lower bound. This gives (3.5). The argument applies to real, possibly nondiscrete valuations. $\square$

4 The all-higher-level contact identity

The key estimate is obtained by keeping the anchor fixed while the higher return level varies. Write

$$m_d = (P^{\circ p^d})'(\beta), \quad A_d = v(m_d - 1), \quad \beta \in \Pi_d.$$

We first prove that $m_d - 1$ is nonzero, before using it as a denominator.

Proposition 4.1 (Multiplier and average contacts). *For every $d \geq 1$ and $\beta \in \Pi_d$,*

$$A_d = \delta_{d-1} + \sum_{j=0}^{d-1} (p-1)p^{d-j-1}\delta_j < \infty. \quad (4.1)$$

The value A_d is independent of β . For every $e > d$,

$$\frac{1}{p^e} \sum_{\alpha \in \Pi_e} v(\beta - \alpha) = c_d := \frac{p-1}{p^{d+1}} A_d \geq b_d := dr^3 + r^2 \left(1 + \frac{1}{p}\right). \quad (4.2)$$

Proof. Differentiate the factorization of Lemma 2.2 at level d and evaluate at β . Since $M_d(\beta) = 0$,

$$m_d - 1 = F_{d-1}(\beta)M'_d(\beta)V_d(\beta). \quad (4.3)$$

The first factor on the right is nonzero by the least-period condition, the second by the simple root product for M_d , and the third because it is a unit. Hence $m_d \neq 1$. Taking valuations and using Lemma 3.2 gives

$$\begin{aligned} A_d &= v(F_{d-1}(\beta)) + \sum_{\gamma \in \Pi_d \setminus \{\beta\}} v(\beta - \gamma) \\ &= \delta_{d-1} + \sum_{j=0}^{d-1} (p-1)p^{d-j-1}\delta_j. \end{aligned}$$

The chain-rule expression for m_d is the product of P' over all points of the cycle. A different anchor only permutes its factors, proving independence of A_d .

Now let $e > d$. Both F_e and F_{e-1} vanish at β . Differentiating $F_e = F_{e-1}Q_e$ there gives

$$F'_e(\beta) = F'_{e-1}(\beta)Q_e(\beta). \quad (4.4)$$

Because $P^{\circ p^d}$ fixes β , the chain rule and characteristic p yield

$$F'_e(\beta) = m_d^{p^{e-d}} - 1 = (m_d - 1)^{p^{e-d}}, \quad F'_{e-1}(\beta) = (m_d - 1)^{p^{e-d-1}} \neq 0.$$

The last exponent is one when $e = d + 1$, so that boundary case also allows division in (4.4). We obtain

$$Q_e(\beta) = (m_d - 1)^{(p-1)p^{e-d-1}}. \quad (4.5)$$

Distinct least periods make Π_d and Π_e disjoint. Using $Q_e = M_e V_e$, the unit value of $V_e(\beta)$ and the exact root product for M_e , equation (4.5) becomes

$$\sum_{\alpha \in \Pi_e} v(\beta - \alpha) = (p-1)p^{e-d-1}A_d. \quad (4.6)$$

All summands are finite. Division by the real number p^e proves the equality in (4.2).

Finally Lemma 3.3 and (4.1) give

$$\begin{aligned}\frac{A_d}{p^d} &\geq \frac{r(p^{d-1} + 1)}{p^d} + \frac{r}{p^d} \sum_{j=0}^{d-1} (p-1)p^{d-j-1}(p^j + 1) \\ &= r \left(1 + \frac{1}{p} + d \frac{p-1}{p} \right).\end{aligned}$$

Multiplication by $(p-1)/p = r$ proves the lower bound in (4.2). In particular $c_d \rightarrow +\infty$. $\square$

Corollary 4.2 (Aligned native-orbit coupling). *For every $e > d$, there is a coupling of μ_e and μ_d all of whose support pairs have distance at most $|s|^{c_d}$. Moreover $d_H(\Pi_e, \Pi_d) \leq |s|^{c_d}$.*

Proof. Fix $\beta \in \Pi_d$. One contact in (4.2) is at least its real average. Choose $\alpha \in \Pi_e$ realizing such a contact. Lemma 3.1 gives

$$|P^{\circ i}(\alpha) - P^{\circ i}(\beta)| = |\alpha - \beta| \leq |s|^{c_d} \quad (i \geq 0).$$

The real probability

$$\pi_{e,d} = p^{-e} \sum_{i=0}^{p^e-1} \delta_{(P^{\circ i}(\alpha), P^{\circ i}(\beta))} \quad (4.7)$$

has first marginal μ_e . Each point of Π_d appears p^{e-d} times in the second coordinate, giving marginal μ_d . The same matched iterates cover both cycle sets and prove the Hausdorff bound. No compatible phases across different pairs of levels are required. $\square$

5 Compact supports and convergence of measures

We record the metric argument at the generality used here. It does not require local compactness of the ambient space, and it separates the closure containing every old cycle from the limiting support.

Lemma 5.1 (Exact finite-cycle distances). *Let T be an isometry of a metric space. For finite single T -cycles E, F with uniform probabilities σ_E, σ_F , put $a = \min_{x \in E, y \in F} d(x, y)$. Then*

$$d_H(E, F) = W_1(\sigma_E, \sigma_F) = W_\infty(\sigma_E, \sigma_F) = a. \quad (5.1)$$

Proof. Choose a nearest pair x, y , let the cycle lengths be m, n , and put $L = \text{lcm}(m, n)$. The probability

$$\pi = \frac{1}{L} \sum_{j=0}^{L-1} \delta_{(T^j x, T^j y)}$$

has the required marginals, since each coordinate point occurs L/m or L/n times. Isometry makes every paired distance equal to a . Every pair in $E \times F$ has distance at least a , so no coupling has smaller mean or essential-supremum cost. The iterates also cover both finite sets, proving both directed Hausdorff distances equal to a . $\square$

Proposition 5.2 (Compact limit of uniformly close cycles). *Let (X, d) be a complete ultrametric space, let $T : X \rightarrow X$ be an isometry, and let C_e be a nonempty finite single T -cycle for each $e \geq 1$. Let σ_e be its uniform probability, and suppose*

$$a_{d,e} = \min_{x \in C_d, y \in C_e} d(x, y), \quad \varepsilon_d = \sup_{e > d} a_{d,e} \longrightarrow 0. \quad (5.2)$$

Then $B = \text{cl}_X(\bigcup_e C_e)$ is compact and $T(B) = B$. There is a nonempty compact set $A \subset B$ with $T(A) = A$ such that $C_e \rightarrow A$ in Hausdorff distance. The probabilities σ_e converge weakly on B to an invariant probability σ with $\text{supp } \sigma \subset A$. For every $d \geq 1$,

$$d_H(C_d, A) \leq \varepsilon_d, \quad W_1(\sigma_d, \sigma) \leq W_\infty(\sigma_d, \sigma) \leq \varepsilon_d. \quad (5.3)$$

Proof. Fix $\eta > 0$ and choose d with $\varepsilon_d < \eta$. Lemma 5.1 makes C_d a finite η -net for every later cycle. Adding all earlier cycle points gives a finite net for the entire union. That union is totally bounded, as is its closure B . The latter is complete because it is closed in X . A complete totally bounded metric space is compact: successive finite covers of radii tending to zero provide a Cauchy subsequence of any sequence, and completeness supplies its limit. Thus B is compact.

The map T sends the union of cycles onto itself. By continuity it maps B into B , and its compact image is closed and contains the dense union. Hence $T(B) = B$. No surjectivity on X was assumed.

By (5.1), the cycle sets are Hausdorff-Cauchy. Define their nonempty compact tail intersection by

$$A = \bigcap_{N \geq 1} \text{cl}_B \left(\bigcup_{e \geq N} C_e \right). \quad (5.4)$$

We verify Hausdorff convergence directly. Given $\eta > 0$, choose N so that $d_H(C_e, C_f) < \eta/2$ whenever $e, f \geq N$. Fix $e \geq N$ and $x \in C_e$. Points in arbitrarily late C_f can be chosen within $\eta/2$ of x ; a subsequence converges in B to a point of A at distance at most $\eta/2$ from x . Conversely, each $y \in A$ is the limit of points in cycles with indices tending to infinity. Choose nearby points of the fixed compact set C_e and pass to a convergent subsequence there. This gives a point of C_e within $\eta/2$ of y . Both bounds are uniform, proving $d_H(C_e, A) \rightarrow 0$. The same argument with d fixed and $d_H(C_d, C_e) \leq \varepsilon_d$ for all $e > d$ gives the first inequality in (5.3).

Continuity implies $T(A) \subset A$. For the reverse inclusion, write $y \in A$ as a limit of $y_j \in C_{e_j}$, $e_j \rightarrow \infty$. Choose a predecessor $x_j \in C_{e_j}$ with $T(x_j) = y_j$. A convergent subsequence of x_j has limit $x \in A$, and $T(x) = y$. Therefore $T(A) = A$.

For $\varphi \in C(B, \mathbb{R})$ let

$$\omega_\varphi(t) = \sup\{|\varphi(x) - \varphi(y)| : x, y \in B, d(x, y) \leq t\}.$$

Compactness gives uniform continuity, so $\omega_\varphi(t) \rightarrow 0$ as $t \downarrow 0$. The coupling in Lemma 5.1 yields

$$\left| \int \varphi d\sigma_e - \int \varphi d\sigma_d \right| \leq \omega_\varphi(\varepsilon_d) \quad (e > d). \quad (5.5)$$

Thus the integrals converge for every such φ . Their limits define a positive norm-one linear functional taking the constant one to one. The Riesz representation theorem on the compact Hausdorff space B gives a regular Borel probability σ . This is convergence of the full sequence. Applying it to $\varphi \circ T$ proves invariance. The continuous nonnegative function $x \mapsto d(x, A)$ has integrals tending to zero by Hausdorff convergence; hence $\text{supp } \sigma \subset A$.

For the transport bound, fix d and consider the finite-cycle couplings with σ_e for $e > d$. Probabilities on the compact metric space $B \times B$ admit a weakly convergent subsequence. One proof takes a countable uniformly dense subset of continuous functions, extracts a diagonal subsequence of their integrals, and uses Riesz representation for the resulting positive functional. The limiting coupling has marginals σ_d, σ . The nonnegative continuous function $(x, y) \mapsto \max\{d(x, y) - \varepsilon_d, 0\}$ has integral zero along these couplings and in their limit. The limit is therefore supported on $d(x, y) \leq \varepsilon_d$. This proves the remaining inequalities in (5.3). $\square$

5.1 Application to the optimal cycles

Apply Proposition 5.2 to the complete disk $\mathcal{D}$, the isometry P , and the cycles Π_e . Corollary 4.2 gives

$$\varepsilon_d \leq |s|^{c_d} \leq |s|^{dr^3+r^2(1+1/p)} \longrightarrow 0.$$

It follows that $C = \text{cl}_K(\bigcup_e \Pi_e)$ is compact, that the full sequence has Hausdorff limit $\mathcal{A}$, and that μ_e has an invariant weak limit μ on C with $\text{supp } \mu \subset \mathcal{A}$. The norm is continuous, so C and $\mathcal{A}$ remain on $|z| = |s|^r$. The inequalities (1.5) have also been proved.

To interpret the limit on $\mathbb{P}_K^{1,\text{an}}$, send $a \in K$ to the evaluation seminorm at a . This map is continuous: each defining coordinate $a \mapsto |H(a)|$, $H \in K[z]$, is continuous. The Berkovich projective line is Hausdorff; see the seminorm and analytification background in [1, Sections 2.3–2.4]. The restriction to compact C is therefore a homeomorphism onto a closed compact image. Pushing μ forward gives a Radon probability on $\mathbb{P}_K^{1,\text{an}}$. Every continuous real test function on $\mathbb{P}_K^{1,\text{an}}$ restricts to one on C , so convergence on C proves the required weak convergence on $\mathbb{P}_K^{1,\text{an}}$. Neither compactness of an ambient classical disk nor metrizable of the entire Berkovich line has been used.

6 Aperiodicity and the Haar adding-machine limit

We first establish the general structure of the compact limit. The polynomial contact identity will then exclude the finite possibilities in that structure theorem.

Proposition 6.1 (Finite cyclic quotients). *Under the hypotheses of Proposition 5.2, $T|_A$ is minimal and uniquely ergodic, and $\text{supp } \sigma = A$. If $|C_e| = p^e$ for a prime p , the system is conjugate either to addition by one on $\mathbb{Z}/p^k\mathbb{Z}$ for some $k \geq 0$, or to addition by one on $\mathbb{Z}_p$. Its invariant probability is the corresponding Haar probability.*

Proof. Fix $x, y \in A$ and $\eta > 0$. Choose a cycle C_e with $d_H(C_e, A) < \eta$ and points $u, v \in C_e$ satisfying $d(u, x), d(v, y) < \eta$. Some $j \geq 0$ has $T^j u = v$. Isometry and the ultrametric inequality give

$$d(T^j x, y) \leq \max\{d(T^j x, T^j u), d(v, y)\} < \eta.$$

Every forward orbit in A is dense. The support of the invariant probability is a nonempty closed forward-invariant subset of A , so it equals A .

Choose a decreasing sequence $\eta_j > 0$ tending to zero. On A define $x \sim_j y$ by $d(x, y) < \eta_j$. The ultrametric inequality makes this an equivalence relation. Its classes are clopen and, by compactness, form a finite quotient Q_j . Since $T(A) = A$ and T is an isometry, it induces a permutation of Q_j . Minimality forces this permutation to be one cycle. Put $q_j = |Q_j|$.

Any invariant probability on A must assign each cell mass $1/q_j$. These partitions refine and have mesh tending to zero. Functions constant on their cells approximate every continuous function uniformly. Hence those masses determine all continuous-function integrals, proving uniqueness of the invariant probability.

Assume now $|C_e| = p^e$. For a fixed j , choose e with $d_H(C_e, A) < \eta_j$. Map $u \in C_e$ to the class of a nearby $x \in A$ with $d(u, x) < \eta_j$. Two choices of x have distance less than η_j , so the map is well-defined. The other directed Hausdorff bound makes it onto, and isometry makes it equivariant. It maps a p^e -cycle onto a q_j -cycle, so $q_j \mid p^e$. Thus $q_j = p^{k_j}$ for some integer $k_j \geq 0$.

Choose a basepoint $a \in A$ and label the cell containing $T^m a$ by $m \bmod q_j$. Transitivity makes this a well-defined labeling of every cell. Refinement is then the reduction map

$$\mathbb{Z}/q_{j+1}\mathbb{Z} \longrightarrow \mathbb{Z}/q_j\mathbb{Z}, \quad q_j \mid q_{j+1}.$$

The resulting coding map

$$A \longrightarrow \varprojlim_j \mathbb{Z}/q_j\mathbb{Z} \quad (6.1)$$

is continuous and injective because the cell diameters tend to zero. A compatible sequence of cells is nested, nonempty, and compact; its intersection is nonempty by compactness and is a singleton by vanishing diameter. This proves surjectivity. The map is a homeomorphism and conjugates T to addition by one.

If k_j is bounded, it eventually stabilizes and the inverse limit is one finite p -power cycle, including a singleton. Otherwise p^{k_j} is cofinal among all powers of p , and (6.1) is $\mathbb{Z}_p$. In each case the invariant probability is uniform on every finite quotient and hence is Haar probability. In the infinite case, the mass of a point is at most p^{-k_j} for every j , proving non-atomicity. $\square$

Proposition 6.2 (Separation from every old cycle). *For every $d \geq 1$, define*

$$U_d = p^d c_d - (p^d - 1)r < \infty. \quad (6.2)$$

For $e > d$, $\beta \in \Pi_d$, and $\alpha \in \Pi_e$,

$$v(\beta - \alpha) \leq U_d, \quad |\beta - \alpha| \geq |s|^{U_d} > 0. \quad (6.3)$$

Consequently $\mathcal{A} \cap \Pi_d = \emptyset$ for every d , and $P|_{\mathcal{A}}$ is the infinite p -adic adding machine.

Proof. Put $h = v(\beta - \alpha)$. The p^{e-d} distinct points

$$P^{\circ j p^d}(\alpha), \quad 0 \leq j < p^{e-d},$$

all have contact h with β : the iterate fixes β and is an isometry. Every remaining point of Π_e has contact at least r , since both points lie on the sphere of radius $|s|^r$. The exact mean identity (4.2) therefore gives

$$c_d \geq p^{-d}h + (1 - p^{-d})r.$$

Rearranging proves (6.3). Its distance bound is uniform in every $e > d$ and every point of the two cycles. Pass to the Hausdorff limit of the higher cycles to obtain $\mathcal{A} \cap \Pi_d = \emptyset$.

Proposition 6.1 leaves only a finite p^k -cycle or the p -adic adding machine. A finite alternative with $k \geq 1$ would lie in the open unit disk and, by uniqueness of the cycle of that period, equal Π_k . This contradicts the separation just proved. A singleton alternative would be a fixed point of P . The fixed points are 0 and $-s$, and neither lies on $|z| = |s|^r$, since $0 < |s| < |s|^r$. The finite alternatives are therefore excluded. $\square$

Completion of Theorem 1.1. Proposition 4.1 gives the finite, anchor-independent contact identity and its growing lower bound. Proposition 5.2 and its quadratic application give compactness, Hausdorff convergence, the entire weakly convergent measure sequence, and (1.5). Propositions 6.1 and 6.2 identify the limit system with $(\mathbb{Z}_p, +1)$ and its measure with nonatomic Haar probability. The support equals $\mathcal{A}$ by minimality. Its embedding in $\mathbb{P}_K^{1,\text{an}}$ is compact and consists of classical points, so the Berkovich support is exactly that same type-I set. $\square$

The finite upper bound (6.2) plays a different role from the growing lower bound for c_d . The latter makes high cycles uniformly close; the former keeps their limit away from each fixed old cycle. Growing approximating periods alone would not exclude a finite limit.

Object	Established structure	Role of the estimate
$C = \text{cl}_K \bigcup_e \Pi_e$	Compact classical invariant set containing all old cycles	Uniform high-level contacts give finite nets at every scale
$\mathcal{A} = \lim_e \Pi_e$	Hausdorff limit, disjoint from every old cycle, conjugate to $\mathbb{Z}_p$	Lower contacts give convergence; the finite average gives separation
μ_e and μ	Full-sequence convergence to the unique invariant probability on $\mathcal{A}$	Aligned orbit couplings give $W_\infty(\mu_d, \mu) \leq s ^{c_d}$

Table 1: Exact implications for the containing closure, the limiting system, and the cycle measures. Unique ergodicity belongs to $\mathcal{A}$, not to C . All distances are classical absolute-value distances.

7 Consequences and scope

The theorem describes two distinct compact invariant sets and their associated measures. Keeping them separate prevents a false unique-ergodicity assertion on a space containing finite cycles.

For any real continuous function φ on C , the proof also gives the explicit modulus-of-continuity estimate

$$\left| \int \varphi d\mu_d - \int \varphi d\mu \right| \leq \omega_\varphi(|s|^{c_d}).$$

In particular a Lipschitz test function of constant L has error at most $L|s|^{c_d}$. The bound is for each fixed allowed multiplier; neither its optimality nor uniformity near the boundary $|s| = 1$ is asserted.

The proof uses the source’s optimal-cycle geometry, coprime Hensel factorization, derivative products, and the elementary characteristic- p substitution identity. Its specific distribution step is the all-anchor contact average with a bound diverging in the anchor level. Compactness and the cyclic inverse-limit argument then identify the resulting measure. No Galois irreducibility, oriented character, or splitting-field nesting conclusion is needed or obtained here. The adding-machine conjugacy is topological, not an asserted analytic linearization.

The theorem does not address characteristic zero, characteristic two, all normalized germs, or counting all periodic points of a polynomial. The comparison with existing measure and ramification results is bounded to the specified sources and does not certify worldwide priority or the present status of every later or unpublished work. The assertion proved here is the positive answer to the explicitly identified version of the selected-cycle question.

Preparation and verification statement. This anonymous manuscript was prepared with AI assistance. The underlying proof arguments received nonauthor review within the same research team; this is internal verification, not external peer review or journal acceptance. No numerical experiment is used as evidence for the theorem. All substantive contact, compactness, and aperiodicity arguments used in the article are included above.

References

- [1] Matthew Baker. An introduction to Berkovich analytic spaces and non-archimedean potential theory on curves. Arizona Winter School lecture notes, 2007. Sections 1 and 2.3–2.4. <https://swc-math.github.io/aws/2007/BakerNotesMarch21.pdf>.

- [2] Steven Hurder and Olga Lukina. Essential holonomy of Cantor actions. *Journal of the Mathematical Society of Japan*, 77(1):57–74, 2025. The background in Sections 2.2–2.3 was checked in arXiv:2205.06285v2, 20 January 2023. <https://arxiv.org/abs/2205.06285v2>.
- [3] Kenneth Jacobs. Equidistribution of the crucial measures in non-Archimedean dynamics. *Journal of Number Theory*, 180:86–138, 2017. Theorem 2 and the measure definition are cited from the PDF of arXiv:1409.4808v2, 28 April 2017. <https://arxiv.org/pdf/1409.4808v2>.
- [4] Karl-Olof Lindahl and Juan Rivera-Letelier. Optimal cycles in ultrametric dynamics and minimally ramified power series. *Compositio Mathematica*, 152(1):187–222, 2016. Question and theorem numbering refer to arXiv:1311.4478v3, 26 May 2015. <https://arxiv.org/abs/1311.4478v3>.
- [5] Jonas Nordqvist and Juan Rivera-Letelier. Residue fixed point index and wildly ramified power series. *Journal of the London Mathematical Society*, 102(2):470–497, 2020. Author version arXiv:1904.04494v3, 3 February 2020. <https://arxiv.org/abs/1904.04494v3>.